Mission 4: Route

The story

"The case is created. Now the driver asks the only question that matters: where do we go? The nearest hospital is not always the right one. A small hospital can scan the brain and give the drug, but only a few can remove a clot. Your fourth mission finds the fastest hospital that can still give this patient the treatment they need."

What you need to know

- Before any treatment, the patient's brain is scanned with a **CT** machine. That is why every hospital that gives the drug also has a CT.
- Each hospital can do some of three things: `CT` (scan), `DRUG` (give the drug), `CLOT` (remove the clot).
- Roads go both ways. The drive time is the same in both directions.
- The ambulance leaves at the time the case was created in Mission 3.

Use these fixed times:

Step	Minutes
Drive to a hospital in the patient's own city	10
From arrival to the drug	30
From arrival to clot removal	60

City names are written without ë and ç (Durres, Korce, Vlore), so every programming language reads them the same way.

The base challenge

Your goal: send each patient to the hospital that gives the best treatment still possible for them, and get them there as fast as the roads allow.

Goal 1: Read the map files

We give you two files in the same format as the atlas. Load both when the program starts. `map load <path>` loads a new version of either file, all or nothing, like `atlas load`.

`data/roads.txt`:

```
# ROAD | city | city | minutes

ROAD | Shkoder     | Lezhe       | 40
ROAD | Lezhe       | Tirana      | 60
ROAD | Lezhe       | Durres      | 60
ROAD | Kukes       | Lezhe       | 100
ROAD | Kukes       | Shkoder     | 165
ROAD | Kukes       | Peshkopi    | 110
ROAD | Peshkopi    | Tirana      | 150
ROAD | Tirana      | Durres      | 40
ROAD | Tirana      | Elbasan     | 50
ROAD | Durres      | Fier        | 70
ROAD | Elbasan     | Fier        | 90
ROAD | Elbasan     | Korce       | 120
ROAD | Fier        | Berat       | 50
ROAD | Fier        | Vlore       | 40
ROAD | Vlore       | Sarande     | 150
ROAD | Fier        | Gjirokaster | 120
ROAD | Gjirokaster | Sarande     | 60
ROAD | Korce       | Gjirokaster | 200
```

`data/hospitals.txt`:

```
# HOSPITAL | name | city | can do (CT, DRUG, CLOT)

HOSPITAL | Qendra Neuro Tirana | Tirana      | CT, DRUG, CLOT
HOSPITAL | Spitali Rozafa      | Shkoder     | CT, DRUG
HOSPITAL | Spitali Adriatik    | Durres      | CT, DRUG
HOSPITAL | Spitali Shkumbin    | Elbasan     | CT, DRUG
HOSPITAL | Spitali Apolonia    | Fier        | CT, DRUG
HOSPITAL | Spitali Kanina      | Vlore       | CT, DRUG
HOSPITAL | Spitali Morava      | Korce       | CT, DRUG
HOSPITAL | Spitali Drino       | Gjirokaster | CT, DRUG
HOSPITAL | Spitali Osumi       | Berat       | CT
HOSPITAL | Spitali Lisus       | Lezhe       | CT
HOSPITAL | Spitali Korabi      | Peshkopi    | CT
HOSPITAL | Spitali Butrint     | Sarande     | CT
```

The hospital names and what they can do are made up for this competition. Kukes has no hospital.

Goal 2: Drive times from a city

`route from <city>` lists every hospital, the shortest drive time to it, and the cities on the way, fastest first. If two hospitals are equally fast, list the one that comes first in `hospitals.txt` first.



```
synapse> route from Fier
HOSPITAL              CITY         CAN DO          DRIVE     ROUTE
Spitali Apolonia      Fier         CT, DRUG        10 min    inside Fier
Spitali Kanina        Vlore        CT, DRUG        40 min    Fier -> Vlore
Spitali Osumi         Berat        CT              50 min    Fier -> Berat
Spitali Adriatik      Durres       CT, DRUG        70 min    Fier -> Durres
Spitali Shkumbin      Elbasan      CT, DRUG        90 min    Fier -> Elbasan
Qendra Neuro Tirana   Tirana       CT, DRUG, CLOT  110 min   Fier -> Durres -> Tirana
Spitali Drino         Gjirokaster  CT, DRUG        120 min   Fier -> Gjirokaster
Spitali Lisus         Lezhe        CT              130 min   Fier -> Durres -> Lezhe
Spitali Rozafa        Shkoder      CT, DRUG        170 min   Fier -> Durres -> Lezhe -> Shkoder
Spitali Butrint       Sarande      CT              180 min   Fier -> Gjirokaster -> Sarande
Spitali Morava        Korce        CT, DRUG        210 min   Fier -> Elbasan -> Korce
Spitali Korabi        Peshkopi     CT              260 min   Fier -> Durres -> Tirana -> Peshkopi
```

Goal 3: Route a case

`route <case id>` reads the case from your database and decides where the ambulance goes. The case already says which treatments are still possible for this patient. Choose with these rules:

- Clot removal is better than the drug alone. The drug is better than a scan alone.
- A hospital counts for a treatment only if it can do it and the patient gets it in time: arrival + 60 min within the clot removal limit, arrival + 30 min within the drug limit.
- Go for the best treatment that some hospital can give in time. Among the hospitals that give it, choose the fastest to reach.
- A patient sent for clot removal also gets the drug there, if arrival + 30 min is still within the drug limit.
- If no hospital can give any treatment in time, go to the fastest hospital with `CT`, for a scan and care.

Save the chosen route in the case.



Example CLI interaction (you may design yours differently):

```
synapse> route P-0001
Case P-0001: Arben Hoxha, Fier, leaves at 08:10
Needs: clot removal (until 07:30 tomorrow), drug (until 12:00)
Fastest hospital that can remove the clot: Qendra Neuro Tirana
  Route:         Fier -> Durres -> Tirana, 110 min
  Arrives:       10:00
  Drug:          10:30
  Clot removal:  11:00
SUCCESS: Route saved for P-0001.

synapse> route P-0002
Case P-0002: Drita Marku, Shkoder, leaves at 06:50
Needs: clot removal (until 22:30). The drug limit has passed.
Fastest hospital that can remove the clot: Qendra Neuro Tirana
  Route:         Shkoder -> Lezhe -> Tirana, 100 min
  Arrives:       08:30
  Clot removal:  09:30
SUCCESS: Route saved for P-0002.

synapse> route P-0003
Case P-0003: Besnik Kola, Kukes, leaves at 10:00
Needs: drug (until 13:30). Clot removal is not possible.
Fastest hospital that can give the drug in time: Spitali Rozafa
  Route:         Kukes -> Lezhe -> Shkoder, 140 min
  Arrives:       12:20
  Drug:          12:50
SUCCESS: Route saved for P-0003.
```

Case P-0004 is created in Mission 3 like the others: Mira Dedja, 61, Korce, time now 09:00, last seen well 06:00, symptoms `cannot-speak, cannot-understand, right-arm-weak, right-face-droop`. That gives severity 8, pipe `left-middle` (big), drug until 10:30.

```
synapse> route P-0004
Case P-0004: Mira Dedja, Korce, leaves at 09:00
Needs: clot removal (until 06:00 tomorrow), drug (until 10:30)
Fastest hospital that can remove the clot: Qendra Neuro Tirana
  Route:         Korce -> Elbasan -> Tirana, 170 min
  Arrives:       11:50
  Drug:          too late (limit 10:30)
  Clot removal:  12:50
SUCCESS: Route saved for P-0004.
```

Example of the saved route inside the case (you may design yours differently):

```
"route": {
  "hospital": "Qendra Neuro Tirana",
  "path": ["Fier", "Durres", "Tirana"],
  "driveMinutes": 110,
  "leaves": "08:10",
  "arrives": "10:00",
  "drugAt": "10:30",
  "clotRemovalAt": "11:00"
}
```




Error handling

Your program must handle and clearly report errors such as:

- "ERROR: roads.txt, line 7: minutes must be a whole number above 0, found "-5"."
- "ERROR: roads.txt, line 12: the road Fier - Vlore is listed twice."
- "ERROR: hospitals.txt, line 4: city "Tirane" is not on any road."
- "ERROR: hospitals.txt, line 9: unknown ability "SCAN". Expected CT, DRUG or CLOT."
- "ERROR: hospitals.txt, line 3: a hospital that gives the DRUG must also have CT."
- "ERROR: No case called "P-0099"."
- "ERROR: The city of case P-0005, "Fierr", is not on the map."
- "ERROR: No road leads from Kukes to any hospital."

The bonus challenge

For extra points.

Bonus 1: Drug first, then transfer

For a patient who needs clot removal, there is a second plan: drive to the fastest hospital that can give the drug, give it, then transfer the patient to a clot removal hospital. Extra fixed times for this plan:

Step	Minutes
Transfer leaves the first hospital, after arrival	45
From arrival at the second hospital to clot removal (the team is ready)	30

Compare both plans and choose with this rule: if only one plan gives the drug in time, choose it. Otherwise choose the plan with the earlier clot removal.

```
synapse> route P-0004
Case P-0004: Mira Dedja, Korce, leaves at 09:00
Plan A, straight to clot removal: Qendra Neuro Tirana
  Arrives 11:50 · drug too late · clot removal 12:50
Plan B, drug first, then transfer:
  Spitali Morava (inside Korce): arrives 09:10 · drug 09:40
  Transfer leaves 09:55 -> Korce -> Elbasan -> Tirana, 170 min
  Qendra Neuro Tirana: arrives 12:45 · clot removal 13:15
Chosen: Plan B. Only this plan gives the drug in time.
SUCCESS: Route saved for P-0004.
```

For Arben (P-0001) both plans give the drug in time, and Plan A removes the clot 25 min sooner (11:00 against 11:25), so Plan A is chosen.

Bonus 2: Coverage report

`map coverage` shows, for every city, the drive time to the nearest hospital that gives the drug and to the nearest one that removes clots. Mark a city `FAR` when clot removal is more than 120 min away.



```
synapse> map coverage
CITY          NEAREST DRUG          NEAREST CLOT REMOVAL
Shkoder       10 min (Shkoder)      100 min (Tirana)
Lezhe         40 min (Shkoder)      60 min (Tirana)
Kukes         140 min (Shkoder)     160 min (Tirana)   FAR
...
Sarande       60 min (Gjirokaster)  290 min (Tirana)   FAR
7 of 13 cities are more than 120 min from clot removal.
```

Bonus 3: What if?

A health planner asks what would change if one hospital could do more. `route whatif <city> <ability>` recomputes the drive to clot removal for every saved case where clot removal is possible, as if the hospital in that city had the new ability. Nothing is written to `hospitals.txt`.

```
synapse> route whatif Korce CLOT
What if Spitali Morava (Korce) could also remove clots?
CASE     CITY      NOW       WITH THE CHANGE   SAVED
P-0001   Fier      110 min   110 min           0 min
P-0002   Shkoder   100 min   100 min           0 min
P-0004   Korce     170 min   10 min            160 min
Total saved: 160 min over 3 cases.
```

Hints

- The direct road is not always the fastest. Look at how the ambulance gets from Fier to Tirana in the example.
- Check a few answers by hand on the map before you trust your program. Try a city far from everything, such as Kukes or Sarande.
- Bonus 2 asks the same question for every city, so your program must stay fast.

What the judges will test

- `route from` every city, compared with the judges' answer sheet.
- The four cases above, then hidden ones, including a time exactly at a limit.
- A hidden `roads-v2.txt` with a closed road and a hidden `hospitals-v2.txt` with a new ability, both with no code changes.
- A city with no road to any hospital.

What this mission saves

The chosen route inside each case. Mission 5 reads the arrival time of every patient who goes to Qendra Neuro Tirana. Mission 7 prints the route.